

No	Roll Number	Project Title	Project Description
1	R324ZRA01, R324ZRA02,R3 24ZRA03	Adaptive Time- Quantum Round Robin Scheduler	time quantum based on process behavior and workload characteristics. This approach reduces context-switch overhead, improves response time, and enhances overall scheduling efficiency.
2	R324ZRA04, R324ZRA05,R3 24ZRA06	Dynamic Memory Management Visualizer	such as paging, segmentation, and virtual memory. Users can observe memory allocation, deallocation, and address translation to better understand memory utilization strategies.
3	R324ZRA07, R324ZRA08,R3 24ZRA09	Virtual Memory Optimization Simulator	Design a simulator that demonstrates demand paging, page faults, memory allocation, and fragmentation under different workloads. The project evaluates how virtual memory mechanisms impact system performance and resource utilization.
4	R324ZRA10, R324ZRA11,R3 24ZRA12	Page Replacement Algorithm Simulator	Build a simulator to implement and compare FIFO, LRU, LFU, and Optimal page replacement algorithms. The system analyzes page fault rates and hit ratios, enabling users to understand the efficiency of different replacement strategies.
5	R324ZRA13, R324ZRA14,R3 24ZRA15	Memory Fragmentation Analysis Tool	Create a system that analyzes internal and external fragmentation caused by dynamic memory allocation. The tool visually presents fragmentation patterns and helps evaluate memory utilization efficiency over time.
6	R324ZRA16, R324ZRA17,R3 24ZRA18	TLB Performance Analyzer	Simulate the behavior of the Translation Lookaside Buffer to study its impact on memory access time. The analyzer evaluates TLB hit and miss rates and demonstrates how caching improves overall system performance.
7	R324ZRA19, R324ZRA20,R3 24ZRA21	NUMA Memory Allocation Simulator	Develop a simulator for Non-Uniform Memory Access architectures to study memory locality and access latency. The project demonstrates how memory placement affects performance in distributed-memory multiprocessor systems.
8	R324ZRA22, R324ZRA23,R3 24ZRA24	Secure Memory Isolation System	Design a system that enforces strict memory isolation between processes to prevent unauthorized access and data leakage. The project demonstrates protection mechanisms that enhance system security and stability.
9	R324ZRA25, R324ZRA26,R3 24ZRA27	Garbage Collection Algorithm Visualizer	Create a visualization tool that demonstrates garbage collection techniques such as reference counting, mark-and-sweep, and generational collection. The system helps users understand automatic memory management and performance trade-offs.
10	R324ZRA28, R324ZRA29,R3 24ZRA30	Deadlock Detection and Recovery Toolkit	Develop a toolkit that detects deadlocks using resource allocation graphs and system state analysis. The toolkit applies recovery techniques such as process termination or resource preemption to restore system stability.
11	R324ZRA31, R324ZRA32,R3 24ZRA33	Banker's Algorithm Simulator	Design an interactive simulator for Banker's Algorithm that demonstrates safe, unsafe, and deadlock-free states. Users can allocate resources dynamically and analyze system safety conditions.
12	12304783, 12404311, 12317374	Automated Deadlock Prediction System	Build a system that predicts potential deadlocks by analyzing process dependencies and resource request patterns. The project aims to prevent deadlocks before they occur through proactive detection strategies.
13	R324ZRB34, R324ZRB35,R3 24ZRB36	Resource Allocation Graph Visualizer	Develop a graphical tool that visualizes process-resource interactions using allocation graphs. The system helps identify circular wait conditions and understand how deadlocks form in operating systems.
14	R324ZRB37, R324ZRB38,R3 24ZRB39	Fair Resource Allocation System	Create a resource allocation system that dynamically adjusts priorities to ensure fairness among processes. The system prevents starvation and improves overall resource utilization in multiprogramming environments.
15	R324ZRB40, R324ZRB41,R3 24ZRB42	Cloud Resource Allocation Simulator	Simulate CPU and memory allocation in cloud computing environments, focusing on efficient workload distribution among virtual machines and optimizing resource usage under varying demand.
16	R324ZRB43, R324ZRB44,R3 24ZRB45	Inter-Process Communication Debugger	Design a debugging tool that simulates inter-process communication mechanisms such as pipes, message queues, and shared memory. The system helps identify synchronization issues and data-sharing bottlenecks.
17	R324ZRB46, R324ZRB47,R3 24ZRB48	Thread Synchronization Visualizer	Develop a visualization tool that demonstrates thread synchronization mechanisms including semaphores, mutexes, and monitors. The project helps users understand concurrency control in multithreaded systems.
18	R324ZRB49, R324ZRB50,R3 24ZRB51	Energy-Efficient CPU Scheduling Algorithm	Design a power-aware CPU scheduling algorithm that reduces energy consumption using Dynamic Voltage and Frequency Scaling and thermal-aware scheduling. The system balances performance, fairness, and power efficiency, making it suitable for mobile, embedded, and battery-powered computing environments.

19	R324ZRB53, R324ZRB54,R3 24ZRB55	AI-Based Predictive CPU Scheduler	Develop an intelligent CPU scheduler that uses machine learning techniques to predict process burst times based on historical execution patterns. The scheduler aims to reduce average waiting time, improve response time, and enhance overall system performance under varying workloads.
20	R324ZRB56, R324ZRB57,R3 24ZRB58	Multi-Core Load Balancing Scheduler	Create a scheduler that dynamically distributes processes across multiple CPU cores to balance workload efficiently. The system monitors core utilization, prevents processor starvation, reduces load imbalance, and improves throughput and responsiveness in multi-core operating systems.
21	R324ZRB59, R324ZRB60,R3 24ZRB61	Real-Time Scheduling Algorithm Simulator	Build a simulator for real-time scheduling algorithms such as Rate Monotonic and Earliest Deadline First. The tool visualizes task execution, deadlines, preemption, and missed deadlines, helping users understand real-time constraints and scheduling behavior.
22	R324ZRB62, R324ZRB63,R3 24ZRB64	Priority Inversion Handling System	Design a system that demonstrates priority inversion problems in real-time systems and implements solutions such as priority inheritance and priority ceiling protocols. The project helps analyze scheduling anomalies and ensures predictable task execution.
23	R324ZRB65, R324ZRB66	Process Lifecycle Visualization Tool	Develop a graphical visualization tool that illustrates the complete lifecycle of a process, including creation, ready, running, waiting, and terminated states. The system highlights context switching, scheduling decisions, and resource usage in an operating system.
24	R324ZRB67,R3 24ZRB68	Intelligent CPU Scheduler Simulator	Develop a simulator for CPU scheduling algorithms (FCFS, SJF, Round Robin, Priority Scheduling) with real-time visualizations. The simulator should allow users to input processes with arrival times, burst times, and priorities and visualize Gantt charts and performance metrics like average waiting time and turnaround time.